

PROJECT TITAN

Product Requirements Document

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Author	Michael Lozada-Longog
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EDUCATIONAL USE NOTICE

Project TITAN is an educational engineering design project. It is not a clinical device, regulatory submission, or product approved for human use.

Document Control

Revision	Date	Description	Author
A	July 2026	Enhanced product requirements baseline with scope, design inputs/outputs, verification links, assumptions, and regulatory-awareness boundaries.	Michael Lozada-Longog

1. Purpose

This Product Requirements Document establishes the baseline design intent, measurable requirements, constraints, assumptions, and verification approach for the Project TITAN mechanical support assembly. All later TITAN documents should trace their decisions and evidence to this baseline.

2. Project Scope

2.1 Included

- Parametric mechanical CAD assembly
- Static structural finite element analysis
- Engineering drawings and bill of materials
- Material selection and manufacturing planning
- Educational risk analysis and prototype planning

2.2 Excluded

- Human-subject testing or clinical claims
- FDA submission or ISO certification
- Dynamic gait, fatigue, biocompatibility, and long-term wear validation
- Commercial production release

3. System Context

User need -> requirements -> architecture -> CAD/design outputs -> analytical and FEA verification -> manufacturing documentation -> future prototype and physical testing.

4. Design Inputs and Outputs

Design Input	Design Output	Current Evidence
Support radial loading through a bearing-supported shaft	Two-bearing shaft subassembly integrated between structural plates	Fusion 360 assembly and load setup
Minimum static safety factor ≥ 2.0	FEA result set	Minimum reported safety factor 2.377
Controlled deformation under	Stiff plate/frame assembly	Maximum reported displacement

static load		0.004 in
Manufacturable custom parts	Dimensioned part drawings	Completed drawing package
Modular and serviceable construction	Separate plates, frames, shaft, bearings, and spacers	Exploded assembly and component tree
Corrosion-resistant lightweight baseline	6061-T6 aluminum design intent	Material trade study; Fusion material assignment requires final audit

5. Requirements

ID	Requirement	Verification	Status
FR-001	Support a rotating shaft using two bearing elements	CAD/assembly inspection	Verified
FR-002	Maintain an interference-free modular assembly	CAD interference and visual review	Verified
FR-003	Allow individual structural components to be replaced	Assembly architecture review	Verified by design
PR-001	Achieve minimum static safety factor ≥ 2.0 in the defined study	FEA result	Verified: 2.377 reported
PR-002	Keep maximum equivalent stress below the material model yield limit	FEA result and material audit	Conditionally verified; material assignment audit open
PR-003	Keep maximum total displacement ≤ 0.005 in under the defined study	FEA result	Verified: 0.004 in reported
MR-001	Use conventional CNC milling/turning for custom metal parts	Drawing/manufacturing review	Verified by plan
MR-002	Use commercially available bearing and hardware elements	BOM review	Verified conceptually
DR-001	Maintain individual part drawings, assembly evidence, report, and controlled documentation	Document review	Verified
VR-001	Retain simulation inputs and assumptions with result evidence	FEA documentation review	Verified with screenshots and report

6. Confirmed Baseline Inputs

Parameter	Value	Basis
Shaft diameter	10 mm	Main-shaft drawing / Measure tool
Shaft length	40 mm	Main-shaft drawing
Bearing center spacing	31 mm	21 mm clear edge gap + two 5 mm radii

Load per bearing	2,000 N	Fusion bearing-load setup
Total applied bearing load	4,000 N	Two loads at 2,000 N each
Side-plate thickness	6 mm	User-confirmed CAD geometry
Upper/lower frame thickness	6 mm	User-confirmed CAD geometry
Maximum von Mises stress	54.947 MPa	Latest refined result
Minimum safety factor	2.377	Latest refined result
Maximum displacement	0.004 in (0.1016 mm)	Latest refined result

7. Assumptions and Limitations

- Static loading only; no fatigue or impact assessment.
- Linear-elastic, isotropic material behavior as configured in Fusion.
- Idealized contacts, constraints, and bearing-load distribution.
- No physical prototype or experimental correlation has been completed.
- The reported Fusion safety factor and stated 6061-T6 design intent are not internally consistent until the final material assignment is audited.

8. Regulatory Awareness Boundary

Future biomedical development would require formal intended-use definition, risk management, human-factors work, verification/validation, and regulatory pathway evaluation. This project uses design-control-inspired organization only; it does not claim compliance with FDA, ISO 13485, or ISO 14971.

9. Acceptance Criteria

Project TITAN V1 is accepted as an educational engineering baseline when the CAD, drawings, FEA, documentation suite, and calculation review are complete; remaining physical-prototype and test activities are explicitly carried as future work.